RAZOR

Live reverse-tape splicing for VCV Rack and Ableton Live



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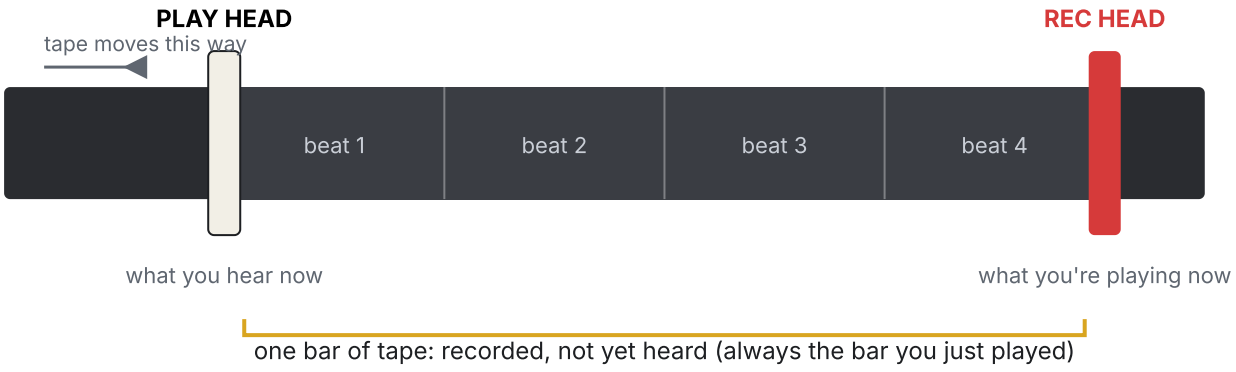
8 Questions and limits

Razor v2.0 · VCV Rack 2 plugin and Max for Live audio effect · source: github.com/hardyfinn/Razor

1 · The idea

On tape, a reverse edit means cutting a piece out, turning it end for end and splicing it back in the same place. Razor does that to a running stream of audio, live.

Most reverse effects flip the audio you have **just heard**, so the material repeats. A tape editor flips the piece that is **about to play**, and the music carries on in time afterwards. To do that live, the "about to play" audio has to exist already. So Razor works like a tape loop with the **record head one bar (4 beats) ahead of the play head**.



Everything you hear is the input exactly one bar late, locked to the clock, so it stays on the grid.

That bar of tape between the heads is the "future" Razor can edit. Press **½, 1, 2 or 4** and that many beats are played backwards from the next beat. Then the play head jumps to where it would have been anyway.

> **One buffer does all four lengths.** Because the gap is always a full bar, the next ½, 1, 2 and 4 beats are always on the tape, ready to cut, whichever button you press.

Why a whole bar?

The lag is fixed at exactly 4 beats, even if you only use the short lengths. It's easier to play against a delay of a whole bar than against part of one: what you hear is always the bar you just played, on the same beat.

What it costs

- Razor's output is **one bar behind its input**. Anything that has to stay in time with it must go through Razor too, or be delayed by a bar.
- Knob moves upstream of Razor are heard **a bar later**.
- Reverses wait for a beat, so Razor needs a clock: the CLK jack or BPM knob in VCV, Live's transport in Live.

2 · The splice: REPLACE and INSERT

REPLACE (the default)

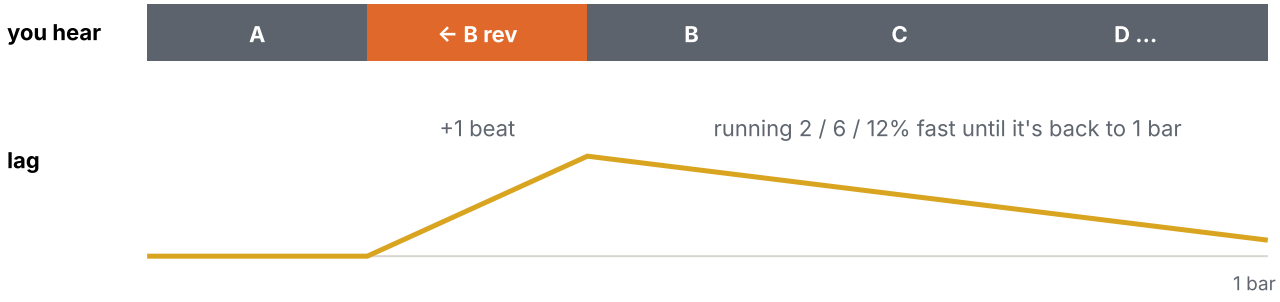
The reversed piece takes the forward piece's place. When it ends, the forward stream comes back exactly where it would have been. Nothing repeats, nothing is lost, and you stay on the grid. This is the real tape edit.



press 1 during A: beat B plays backwards, then C arrives exactly on time

INSERT

The reversed piece is **added**: after it, the tape carries on from the start of B, so the stream falls behind the grid by the length of the piece. Razor then runs the tape slightly fast (varispeed) until it's back to exactly one bar late. You hear a small, gradual pitch rise as it catches up: the tape op's "fix it with varispeed" sound.



The catch-up speed is in the right-click menu (VCV) or the **Insert catch-up** menu (Live): Gentle 2% (about ⅓ semitone), Medium 6% (about 1 semitone) or Fast 12% (about 2 semitones). At 6%, catching up a 1-beat insert takes about 17 beats.

> If an insert would push the lag past what the tape can hold, that piece is spliced as REPLACE instead.

Back to back

Press a length while a reverse is playing and it's queued: it starts the moment the current piece ends (in INSERT mode, on the next beat after it). The light blinks while a press is waiting and stays on while it plays.

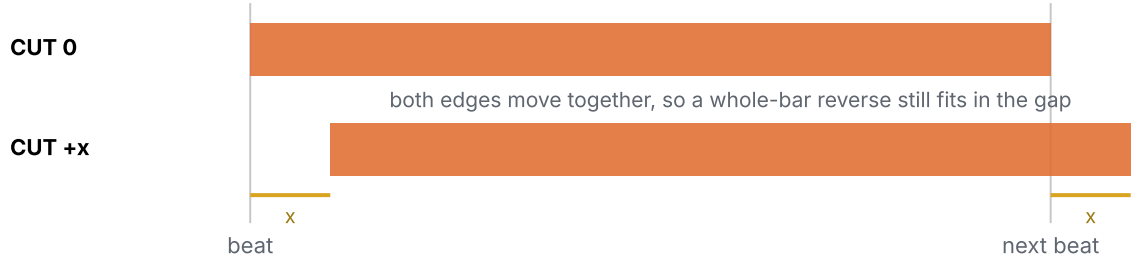
3 · Shaping the cut: CUT, SNAP and ANGLE

Why reversed drums can feel off the beat

With the cuts exactly on the grid, each kick's sharp hit still lands on a beat after the flip. But the flip turns the hit into the **end** of the sound: you hear its tail first as a swell, and the punch at the end. The energy arrives in the wrong part of the beat, even though the peak is technically on the grid. Kicks that start a few milliseconds after the clock, which is common with analogue drums, get pushed further off.

CUT: move the razor

On real tape the splice point *is* the cut, so moving the razor moves both. CUT slides **both edges** of the piece earlier or later than the grid, up to ⅛ beat, in the tape and in when you hear it. The forward stream still comes back on the grid.



- Turning CUT by **x moves the reversed hits by 2x**: it's a sensitive control, and small moves go a long way.
- **CW (later)** pushes the reversed hits later.
- **CCW (earlier)** pulls them earlier, so the swell starts sooner and peaks before the beat.

SNAP: let Razor find the hits

Because Razor always holds a whole bar that you haven't heard yet, it can look at it first. With SNAP on, it finds the hit nearest each cut edge (within ⅛ beat) and puts the razor **2 ms before it**. Cuts never land in the middle of a kick, and each reverse is placed by its own hits, without you setting the knob. CUT then becomes a fine offset from the snapped point. A tool that can only see the past can't do this.

ANGLE: the splice crossfade

A razor block has slots at different angles. A 90° cut is a hard butt splice; an angled cut crosses the tape over a short time, which works like a crossfade. ANGLE uses the real numbers for ¼" tape running at 15 inches per second:



90°
0.2 ms · butt splice, clicky

45°
about 17 ms · the classic cut

22.5°
about 40 ms · soft

45° is the default and is marked on both panels. The crossfade starts at each edge, so the timing of the splice doesn't move.

4 · The VCV module

28 HP. Drop `Razor-2.0.0-mac-arm64.vcvplugin` into your Rack 2 plugins folder and restart Rack. Patch audio into IN L (and IN R for stereo), patch a clock into CLK or set BPM, and take OUT L / OUT R to your mixer.

> **Clock:** one pulse per beat by default. If your clock sends 16ths (DROID, Metropolix, Pam's at x4), set *Clock pulses per beat* to 4 in the right-click menu. The BPM on the display shows you straight away if it's wrong.



Controls

Control	What it does
½ / 1 / 2 / 4	Reverse that many beats, starting at the next beat (or bar with BAR). Buttons always fire.
gate jacks	One under each button: the same as pressing it, for DROID, Marbles, a sequencer... Each gate is checked against PROB first.
SNAP	Cut 2 ms before the nearest hit (see section 3).
BAR	Start on the next bar line, not the next beat. Patch RESET so Razor knows where your bar starts.
INSERT	Add the piece, then catch up with varispeed (see section 2).
CUT	Move both cuts up to ⅛ beat earlier (CCW) or later (CW).
ANGLE	Splice crossfade, 90° to 22.5°; yellow mark = 45°.
PROB	The chance that a gate reverses (0–100%). Buttons ignore it.
BPM	Tempo when nothing is patched to CLK.
CV + CV TO	One CV input, aimed at **cut**, **angle** or **prob** with the small CV TO knob. Cut: ±5 V sweeps the whole range. Angle and prob: 0–10 V adds to the knob.

Jacks

Jack	
IN L / IN R	Audio in. Mono: patch L only.
CLK	Clock; the green light flashes on each beat. A big tempo change jumps straight to a bar of the new tempo; small drift is followed with varispeed.
RESET	The next beat is the first of a bar (for BAR).
OUT L / OUT R	The tape: one bar late, with the splices.
REV	10 V while a reversed piece plays. The red light shows the same.

Right-click menu: clock pulses per beat (1, 2, 4, 24), INSERT catch-up speed, and *Clear the tape*.

5 · The Max for Live device



The device layout (mock-up; the display is drawn by the device's own script from the engine's output).

Install

1. Put `Razor.amxd` and `razor_display.js` **in the same folder**, for example `User Library/Presets/Audio Effects/Max Audio Effect/`. The .js file draws the display.
2. Drag `Razor.amxd` onto a track, group or the master. (Optional: open it in Max and press **Freeze Device** to make one file that carries everything.)

Where to put it

Master, group or a bus of the whole mix	Nothing to fix: everything is a bar late together, so you won't notice. This is the easiest way to use it on a finished mix.
A single track	That track lands a bar behind the rest. Either drag its clips one bar earlier, or group it with everything it has to stay in time with and put Razor on the group.

Razor follows Live's tempo and transport by itself. Reverses only fire while Live is playing, because they wait for the next beat. It assumes 4/4.

Keyboard triggering with Key Map

The four big buttons are ordinary Live parameters, so Live's own mapping drives them:

1. Press `Cmd+K` (Mac) or `Ctrl+K` (Windows). Mappable controls turn yellow.
2. Click ½, then press the key you want, for example `1`. Then **1** → `2`, **2** → `3`, **4** → `4`.
3. Press `Cmd/Ctrl+K` again to leave Key Map mode.

`Cmd/Ctrl+M` does the same for MIDI pads or a keyboard. SNAP, BAR, INSERT and Clear tape can be mapped too. The buttons can also be **automated** in the Arrangement, so you can write reverses into the song.

> Live only sends your computer keyboard to mapped controls when it isn't using the keyboard for MIDI. If your keys play notes, turn off **Computer MIDI Keyboard** (the `M` key toggles it).

Controls

No PROB or CV input here: Live's automation, clip envelopes and Follow Actions do that job.

½ / 1 / 2 / 4	Reverse that many beats from the next beat (or bar).
Cut	−100% to +100% = ⅛ beat earlier to ⅛ beat later.
Angle	Splice angle in degrees, 22.5–90. Default 45°.
SNAP · BAR · INSERT	As on the VCV module.
Insert catch-up	Gentle 2%, Medium 6%, Fast 12%.
Clear tape · REV	Silence the whole tape · lights while a piece plays reversed.

6 · Reading the display



Halfway through a 1-beat reverse. Drawn by the Live device's display script from the engine's real output.

Both versions show the same thing: the tape between the heads, scrolling left as it plays.

Status (top left)	**FORWARD**; **ARMED 2 BEATS: next beat** while a press waits; **REVERSING 1 BEAT** while a piece plays (with "(insert)" in INSERT mode). VCV also shows "gate skipped (PROB)".
Top right	The tempo. After an INSERT: "catching up +6.0% +312 ms", meaning how fast the tape is running and how far behind the grid it still is.
White line, left	The **play head**: what you hear. Half a beat of already-heard tape is shown to its left.
Red line, right	The **record head**. Everything between the two lines is the bar you just played.
Vertical lines	Beats; bar lines are brighter.
Yellow outline	Where an armed reverse will cut.
Orange box	The piece being reversed. The play head turns orange and runs backwards through it.
Small top ticks (VCV)	The hits Razor has found; yellow when SNAP is on.

> A handy habit: watch the right-hand end of the strip. The transient you see arriving at the record head is the one you'll be flipping in a bar's time.

7 · Patch and arrangement ideas

Razor is built for loop fatigue: small, timed edits that keep a loop moving without leaving the grid.

1. **Reversed kick fills.** Four-to-the-floor kick into Razor. Hit ½ on beat 4 of every 8th bar. Turn SNAP on and CUT a little CCW so the reversed kick swells *into* the next downbeat. In Live, map ½ to a key and play it by hand, then record it as automation.
2. **A loop that edits itself (VCV).** A DROID random gate or Marbles' t outputs into the gate jacks, PROB around 25%, and a slow random CV into CV set to **cut**. Every pass gets its own edits, and REPLACE always lands back on the grid, so it never drifts. A good starting point for Autechre-style self-mutating patterns.
3. **Tape-op transitions.** INSERT on, catch-up Gentle, and press **4** on the last bar of a section: a whole bar flipped, then the music slides back onto the grid over the next few bars with a slight pitch rise. Works well on the master going into a breakdown.
4. **Section markers.** In Live, put Razor on the master and write a **Rev 2** automation hit two beats before every new section. The reverse flags the change for the listener like an old edit-tape transition.
5. **REV as a modulation source (VCV).** REV into a VCA on a reverb send so only the reversed pieces get washed out. Or REV into a filter's CV for a darker reverse.
6. **Detroit chords, flipped.** Run a Hapax chord-playground pad through Razor and use **1** on the off-beat stabs with ANGLE near 22.5°. The soft splices turn each stab into a swelling reverse pad that resolves on the next beat.
7. **Hand-played edits.** Map ½ / 1 / 2 / 4 to four pads. Treat them like a drummer's fills: play along with the bar you just heard, which is always the one sitting between the heads.

Starting points

Material	SNAP	CUT	ANGLE
Drums, tight	On	0 to slightly CCW	60–90°
Full mix	On	0	45°
Pads, chords	Off	0	22.5–35°
Glitchy edits	Off	CV / random	90°

8 · Questions and limits

Why is everything late?

That's the trick that makes it work: the bar between the heads is what Razor edits. Put it somewhere where a bar of delay doesn't matter (the master or a group in Live, or after everything that needs to stay in sync in VCV).

I pressed a button and nothing happened.

- Razor waits for the next beat (or bar with BAR). The light blinks and the display says ARMED while it waits.
- No clock: in VCV check CLK and the clock setting in the right-click menu; in Live the transport must be playing.
- VCV gates are subject to PROB: at 0% gates never fire, and the display says "gate skipped (PROB)".

The tempo on the VCV display is four times too fast.

Your clock sends 16ths. Set *Clock pulses per beat* to 4 in the right-click menu.

After INSERT the pitch drifts up.

That's the catch-up: the tape runs fast until it's one bar late again. Use Gentle for a barely audible drift, or Fast to get back quickly.

Can I change the lag?

No. It's always exactly one bar, on purpose: it's easiest to play against, and it's long enough for every length.

Time signatures

Razor counts 4 beats to a bar. In Live it follows the tempo but assumes 4/4.

Technical notes

- The tape holds about 24 seconds in VCV and about 43 seconds at 48 kHz in Live: enough for a bar at 40 BPM plus inserts.
- A big tempo change (more than 5%) jumps straight to one bar of the new tempo with a splice crossfade. Smaller drifts are followed with varispeed.
- SNAP's hit detector compares a fast and a slow envelope, so very soft or smeared material may give it nothing to snap to. Razor then cuts on the grid.
- The engines are the same design in both versions: `src/RazorEngine.hpp` (VCV) and `m4l/razor.genexpr` (gen~). Both pass the same 10 offline tests.